Supplementary Material: Decomposing the GP Advantage in Offline Model-Based Optimization

Anonymous submission

This supplement contains the proofs of Propositions 1–2, the full per-method result tables, the identifiability (matched-tuning) control, the β and ensemble-size ablations, the official-baseline cross-check, the significance details, and the complete experimental configuration. It is provided for reproducibility and is not part of the main body.

1 The Nearest Crossed-Design Precedents

The body claims that no prior offline-MBO work reports a two-way surrogate$\times$optimizer decomposition, and names the nearest near-miss rather than leaving it unstated. Table 3 of Tan et al. (2025) is a genuine 9×2 crossed grid in offline MBO with a clean 4×2 optimizer-on-trained-model sub-grid, but it crosses training *loss* against method, not model class, and its method axis bundles model family with search paradigm, so it is not commensurable with ours. One further title names both axes (Elsayed and Lacor 2014) but its primary text was unobtainable, and robust-parameter-design methodology is characteristically sequential, which would fail the crossing requirement.

2 Proofs

Notation. Objective $f : \mathcal{X} \to \mathbb{R}$; surrogate posterior mean μ and uncertainty $\sigma > 0$; LCB $L_\beta(x) = \mu(x) - \beta\sigma(x)$, $\beta \geq 0$. For a distribution Q on $\mathcal{X}$, L_β is a *valid* $(1-\delta)$ *lower bound under* Q if $\Pr_{x\sim Q}(f(x) \geq L_\beta(x)) \geq 1-\delta$.

Proposition 1 (Coverage of the premise is LCB validity)**.** *For any Q, β, δ, $\Pr_{x\sim Q}(f(x) \geq L_\beta(x)) = \Pr_{x\sim Q}(\mu(x) - f(x) \leq \beta\sigma(x))$, so L_β is a valid $(1-\delta)$ lower bound under Q iff the premise $\{\mu - f \leq \beta\sigma\}$ has Q-probability at least $1-\delta$.*

Proof. Since $\sigma > 0$, pointwise $f(x) \geq \mu(x) - \beta\sigma(x) \iff \mu(x) - f(x) \leq \beta\sigma(x)$. The two events coincide as subsets of $\mathcal{X}$ and hence have equal probability under any Q. □

Sanity checks. β=0 gives $\Pr(f \geq \mu) = \Pr(\mu - f \leq 0)$; $\beta \to \infty$ gives coverage $\to 1$. The identity is dimensionless and uses $\sigma > 0$ once.

Proposition 2 (Split-conformal repair; shift-limited transfer)**.** *Let $\{(x_i, f(x_i))\}_{i=1}^n$ be exchangeable from P, $r_i = (\mu(x_i) - f(x_i))/\sigma(x_i)$ the signed one-sided nonconformity, and $\hat{q}$ the $\lceil (n+1)(1-\delta)\rceil$-th smallest of $r_1, \ldots, r_n$. Then for a fresh $x_{n+1} \sim P$ exchangeable with the calibration set, $\Pr(f(x_{n+1}) \geq \mu(x_{n+1}) - \hat{q}\,\sigma(x_{n+1})) \geq 1-\delta$. Under a shifted proposal $\Pi \neq P$ with density ratio $w = d\Pi/dP$, validity is restored by weighting the calibration quantile by w (weighted conformal).*

Proof. The normalized scores $r_1, \ldots, r_{n+1}$ are exchangeable, so by the split-conformal rank argument $\Pr(r_{n+1} \leq \hat{q}) \geq 1-\delta$. The event $r_{n+1} \leq \hat{q}$ implies $\mu(x_{n+1}) - f(x_{n+1}) \leq \hat{q}\,\sigma(x_{n+1})$, i.e. $f(x_{n+1}) \geq \mu - \hat{q}\sigma$; monotonicity of probability gives the bound. The covariate-shift clause is the weighted-exchangeability extension of Tibshirani et al. (2019): replacing the empirical quantile of $\{r_i\}$ with its w-weighted analogue restores marginal validity under Π. At $\Pi = P$, $w \equiv 1$ recovers the unweighted bound. □

The shift clause is stated for completeness and is not applied. The weighted extension requires a *known* density ratio w, and that is the form in which the guarantee holds; the estimated case is supported empirically rather than by a theorem bounding the coverage gap (Tibshirani et al. 2019), and weighted procedures are exact when the magnitude of the shift is known (Angelopoulos and Bates 2023). Here Π is the distribution of designs an optimizer proposes after ascending a surrogate—the setting in which w is least knowable, and arguably not well defined, since the proposals are the deterministic output of an optimization concentrated on a low-dimensional set. We neither estimate w nor apply weighted conformal anywhere in this paper. The clause is recorded because the proposition would be incomplete without it, not because it is available to us.

That unavailability is itself the explanation for a result reported below. Table S9 shows split-conformal repair restoring in-distribution coverage to its 0.90 target on *every* task while leaving out-of-distribution coverage erratic—spanning 0.00 to 1.00 there, with exactly 0.00 on four of the fourteen tasks. That contrast is engine-robust, which is why we rest on it: recomputed on the audited engine the same repair again hits 0.90 on all seven synthetic tasks while OOD coverage again scatters, from 0.11 to 1.00. The floor differs between engines; the pattern does not. Unweighted split conformal is

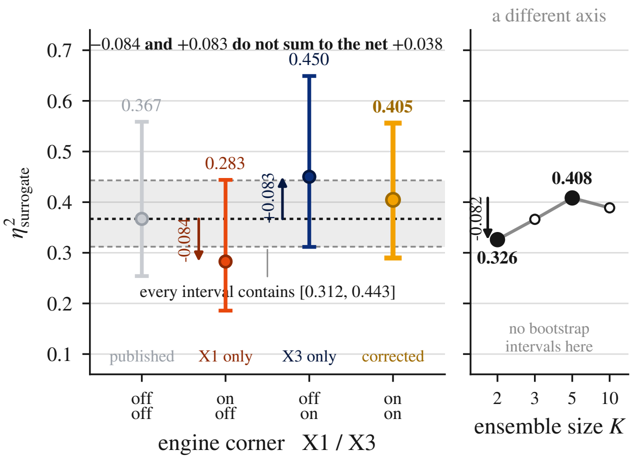

Figure S1: Four-corner confound decomposition (relocated from the main body). The four engine corners are measured levels, not a composing cascade: target scaling alone (X1) moves η^2_{surr} *down* 0.084 and the candidate/oracle rule alone (X3) *up* 0.083, yet the corner where both apply sits 0.038 *above* the uncorrected one, so the corrections are not additive and the corrected 0.405 $[0.290, 0.556]$ is a measured joint level, not a sum. The shaded band is the region every corner's bootstrap interval contains—the four overlap so heavily that no corner is separable from another at $n{=}7$ (the pre-registered KB5 result). The right panel is ensemble size, a confound on a separate axis, not a fifth corner. Synthetic grid, $n{=}7$ tasks, 30 seeds, $B{=}10{,}000$ bootstrap.

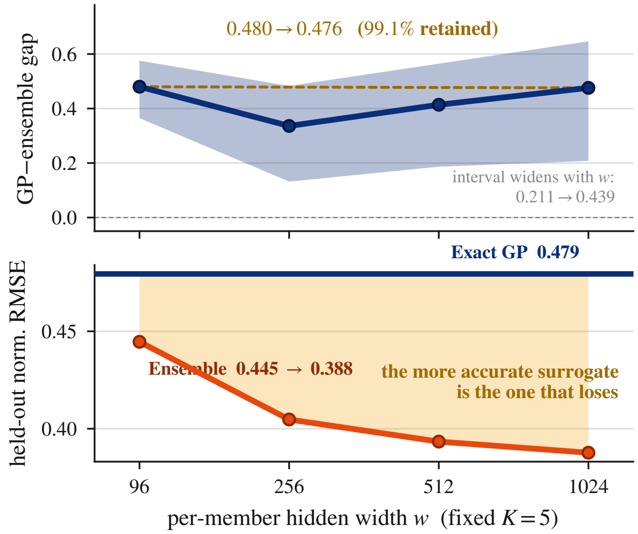

Figure S2: Capacity and accuracy ruled out (Eliminations 2–3; relocated from the main body). Top: at fixed $K{=}5$ the GP–ensemble gap stays flat over a $10.7\times$ width sweep, 0.480 $[0.365, 0.576]$ at $w{=}96$ to 0.476 $[0.208, 0.647]$ at $w{=}1024$ (99.1% retained). The curve is non-monotone (it dips to 0.336 at $w{=}256$ and returns) and the bootstrap band widens with w, so $w{=}1024$ is the least precise point and the supported claim is that the gap *does not close*, not that it is identical there; the sweep stops at 1024 while the kernel limits are asymptotic, so it answers the width objection at practical widths only and the depth-driven form remains open. Bottom: the ensemble's held-out normalized RMSE (0.445, 0.405, 0.393, 0.388 at $w{=}96, 256, 512, 1024$) falls monotonically with width and lies below the exact GP's 0.479 at every width—the more accurate surrogate is the one that loses. Synthetic grid, 7 tasks, 30 seeds, $B{=}10{,}000$; RMSE on a 20% split the grid never saw.

valid under exchangeability, which the optimizer's proposals violate by construction, and the weighting that would repair the shift needs the w we have just said is unavailable. So the erratic OOD column is what the theory predicts rather than an anomaly, and it reinforces the separate finding that premise coverage is separable from optimization outcome: a procedure can be made to cover in-distribution without that touching where the optimizer actually goes.

3 Full Result Tables

Table S1 reports the full synthetic grid including the domain baselines (COMs, CbAS, gradient ascent); Table S2 the corresponding average ranks. The ensemble×gradient and gradient-ascent/COMs rows are the boundary-collapse cells discussed in the body. Both tables are on the *uncorrected* engine, which is stated in each caption and mapped in Section 11: they are the grid as previously published, retained so that the corrections in the body can be read against what they corrected. Every number the body asserts is on the audited engine instead.

4 Identifiability: Matched Tuning (GATE-1)

Exact and sparse GPs can spend per-run marginal-likelihood budget that ensembles cannot. The *matched* arm freezes GP hyperparameter fitting so every surrogate receives the same (zero) per-run tuning budget. The surrogate main effect on synthetic tasks is $\eta^2_{\text{surr}}{=}0.369$ unmatched and 0.275 matched—a 75% retention. Both figures are on the *uncorrected* engine, as in Table S1, rather than the body's audited corner; the control is a within-engine contrast, so what it licenses is the retention ratio and not either level. The unmatched figure is this arm's own estimate on that engine and is not Table S3's 0.367 re-rounded: the two runs differ by 0.002, an estimator offset rather than an engine one, and we print three decimals here so that the pair is not read as a second value of the body's headline. The optimizer main effect stays small in both ($\eta^2_{\text{opt}}{=}0.01 \rightarrow 0.02$) and the interaction stays large ($\eta^2_{\text{inter}}{=}0.17 \rightarrow 0.12$). The GP advantage is therefore a surrogate-class property, not a per-run tuning artifact.

5 Mechanism Controls

The Section-5 mechanism rests on three controls, all on the synthetic tasks (task-normalized, averaged over optimizers).

Pessimism off ($\beta{=}0$). Re-running the full 3×3 grid at $\beta{=}0$ (no σ penalty) and reading both conditions on a single β-invariant normalizer, the GP–ensemble marginal gap

Table S1: Full synthetic grid, 100th-percentile score (30 seeds), including domain baselines, on the *uncorrected* engine—X1 off and X3 off, i.e. the ensemble trains on raw targets and the candidate/oracle protocol is unequalized. **Bold** = best per task. The body reports the audited corner (X1 and X3 on) throughout, so cell values here are not the body's; Section 11 maps every supplementary number to its engine.

Method	Branin	Styblinski	Levy	Rosenbrock	Rastrigin	Ackley	Griewank
Ens+Grad	-9.27	6.34	-2.14	-0.28	-7.71	-3.66	-2592
Ens+Pert	-0.78	33.03	-0.40	-0.12	-8.44	-6.32	-395
Ens+CMA	-14.01	5.15	-3.19	-0.48	-10.81	-4.31	-2613
GP+Grad	$\mathbf{-0.40}$	27.57	$\mathbf{-0.05}$	-0.09	-4.83	$\mathbf{-0.55}$	$\mathbf{-0.94}$
GP+Pert	-0.40	$\mathbf{36.15}$	-0.24	-0.08	-8.28	-6.28	-270
GP+CMA	-0.40	26.65	-0.05	-0.09	-5.06	-0.59	-1.00
SVGP+Grad	-0.45	11.83	-0.08	$\mathbf{-0.04}$	$\mathbf{-2.83}$	-0.70	-2.09
SVGP+Pert	-0.40	34.31	-0.25	-0.08	-8.27	-6.14	-275
SVGP+CMA	-0.53	11.53	-0.08	-0.04	-2.99	-0.74	-2.15
COMs	-9.73	8.21	-4.44	-1.31	-6.97	-4.49	-2078
CbAS	-5.64	33.85	-0.21	-0.05	-8.02	-5.10	-276
Grad. Asc.	-8.35	3.51	-2.43	-0.31	-9.08	-3.04	-2701

Table S2: Average rank over the 7 synthetic tasks (derived from Table S1, so likewise the *uncorrected* X1-off/X3-off engine; lower is better).

Method	Rank	Method	Rank
GP+Grad	2.57	GP+Pert	5.71
SVGP+Grad	3.29	SVGP+Pert	5.86
GP+CMA	3.57	CbAS	6.00
SVGP+CMA	4.29	Ens+Pert	8.14
Ens+Grad	8.57	COMs	9.43
Grad.Asc.	9.86	Ens+CMA	10.71

Table S3: Four-corner decomposition (synthetic, 7 tasks, 30 seeds, B=10,000 bootstrap; relocated from the main body, whose prose reports every entry). X1 standardizes the ensemble's targets; X3 equalizes the candidate/oracle protocol. The intervals overlap heavily: corner differences are *not* resolvable at n=7.

X1 / X3	η^2_{surr} [95% CI]	η^2_{opt}	η^2_{inter}	Friedman p
off / off	0.367 $[0.254, 0.559]$	0.013	0.165	6.1×10^{-5}
on / off	0.283 $[0.186, 0.444]$	0.036	0.146	1.7×10^{-3}
off / on	0.450 $[0.312, 0.649]$	0.006	0.152	4.1×10^{-5}
on / on	$\mathbf{0.405}$ $[0.290, 0.556]$	0.005	0.160	8.1×10^{-4}

is 0.319 $[0.196, 0.460]$ at β=0 against 0.525 $[0.406, 0.614]$ at β=2. Two things follow, and both are asserted. First, 61% of the advantage survives with σ removed entirely and the interval excludes zero, so the advantage rests on a substantial mean-quality base. Second, the paired increment is 0.203 $[0.007, 0.396]$ with p=0.020, which *excludes* zero: pessimism is a significant amplifier of that base, not a bystander. A normalization-free cross-check (per-task z-score by the task's own pooled cell-mean standard deviation) gives 0.904 and 1.473 standard deviations respectively—the same 0.61 ratio—so the conclusion is not an artifact of min–max scaling. Per-task, the ensemble$\times$gradient cell still collapses

Table S4: The five confounds (relocated from the main body, whose §3 prose reports every name, signed effect and kill-criterion outcome). Signed effect is on η^2_{surr} (synthetic, n=7, 30 seeds) except Confound 5, which acts on η^2_{opt}. The two *protocol* confounds move the headline against the ensemble-size confound; the net is the body's §4.

Confound	What it breaks	Signed effect
1. Target scaling	ensemble trains on raw y	$0.367 \to 0.283$
2. Candidate/oracle	two estimands, one column	$0.367 \to 0.450$
3. Ensemble size K	headline fixed at the peak	0.326–0.408
4. β–σ match	per-task, not aggregate	ratio 0.07–1.44
5. Query budget	optimizer = search effort	$0.005 \to 0.038$

without pessimism (Griewank -42.7, Rastrigin -9.5) while the GP does not (Griewank -0.96); these are the audited β=0 grid's own cell means, on the same engine as the interval above.

Two cautions on these numbers. The per-task gaps are heterogeneous: on Branin the β=0 gap (0.664) *exceeds* its β=2 gap (0.468), while on Griewank-30D the β=0 gap is nearly absent (0.098); at n=7 this spread is descriptive only. And the figures above are *not* comparable to gaps computed with a normalizer refit separately at each β, which express the two conditions in different units; such figures may be read within a β condition but never as a cross-β ratio.

Matched subsample. Training the ensemble on the GP's score-biased 800-point subsample rather than all data *widens* the gap to 0.76 (the ensemble marginal falls from 0.34 to 0.08): the GP wins with less data, so the biased subsample is not its edge. This control is computed on the legacy per-condition normalizer and is *not* on the same ruler as the β=0 figures above; only its direction should be read across the two.

Cross-proposal coverage. Table S7 measures each surrogate's premise coverage in-distribution, on its own proposals, and on the other surrogate's proposals. The ensemble premise

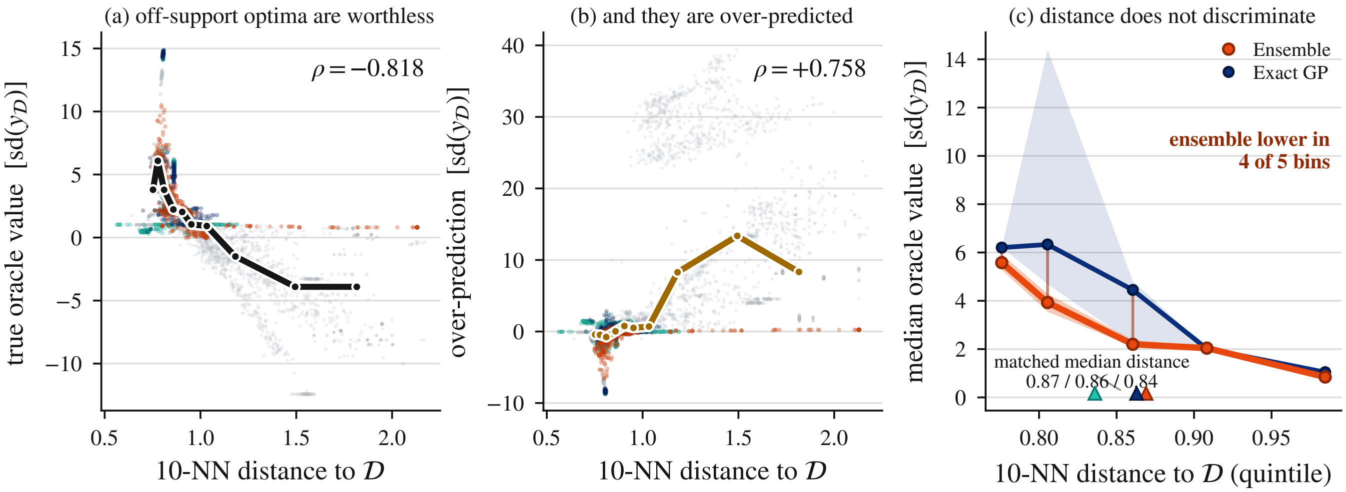

Figure S3: The landscape law is real but does not discriminate (Elimination 7; relocated from the main body and reproduced full-width here, where every label is legible). Distance predicts worthlessness (a, ρ=-0.818) and over-prediction (b, ρ=$+0.758$) across all 5,040 returned optima, yet the surrogate classes are not separated by it (c): at matched median distance (0.87/0.86/0.84) the ensemble is still lower in oracle value, in four of five bins, and the fifth is tied. Panel (c) is the ensemble against the exact GP (SVGP omitted, its distance–value correlation near zero); bands are 95% bootstrap intervals on the bin median. The body reports every number in this figure in prose. Audited engine, X1 and X3 on, β=2, K=5, 30 seeds.

holds in-distribution (0.73) and on the GP's proposals (0.97) but collapses on its own (0.42); the GP premise holds on both (0.97). The ensemble is not miscalibrated in general—only where its own gradient optimizer drives it.

That last sentence points the same way as the interaction reported in the body, and the two are worth reading together—though they are not the same measurement, and we state the difference rather than eliding it. Premise validity here is a property of a *pair* rather than of the surrogate alone: the same ensemble premise holds at 0.97 on points the GP's pipeline proposed and collapses to 0.42 on points its own pipeline proposed. What varies across those two columns is the proposal source—a different surrogate, driven by its own optimizer—so this is a surrogate$\times$proposal-source contrast and not a direct read-out of the surrogate$\times$optimizer interaction the ANOVA estimates. It is corroborating evidence that premise validity is pair-dependent, not a second estimate of η^2_{inter}. Coverage figures are probabilities rather than min–max normalized scores, so the 0.42-versus-0.97 contrast is untouched by the normalizer critique that bears on every η^2 in this paper. As stated in the body, the 0.97 premise-coverage value belongs to the scikit-learn exact GP rather than the differentiable GP the grid scores; the repository carries both implementations.

Simple effects. Once an interaction is significant, main effects can mislead, and the standard remedy for a 3×3 design is to report the simple effects alongside the two main effects. Table S5 gives η^2_{surr} conditional on each optimizer, computed from the stored per-seed data on the *shared omnibus normalizer*. That sharing matters and is easy to get wrong: re-normalizing within a single optimizer's three cells inflates the perturbation figure to roughly 0.73, because min–max rescaling to any three points restores a full unit spread

Table S5: Simple effects: η^2_{surr} conditional on each optimizer, shared omnibus normalizer, all four engine corners. The primary corner is bold.

Corner	at perturbation	at gradient	at CMA
off / off	0.004	0.666	0.814
on / off	0.004	0.388	0.763
off / on	0.019	0.790	0.813
on / on	**0.007**	**0.735**	**0.762**

by construction. The perturbation column sits between 0.004 and 0.020 in every corner against 0.39–0.81 for the two aggressive optimizers, so the asymmetry is corner-invariant in the same way the interaction is. The two X1-off corners retain the target-scaling confound, which is why their intervals are widest.

The same asymmetry in raw oracle units. The simple-effects table is computed on normalized scores. The identical pattern is visible without any normalization at all, which is why we report both: Table S6 gives the gap between the better GP class and the ensemble in raw oracle units, per task and per optimizer, on the audited engine. The perturbation gap is smaller than both aggressive-optimizer gaps on all seven tasks. These are two views of one asymmetry rather than two independent findings—the raw-units form is the interaction expressed in units the normalizer critique cannot reach—and neither view establishes the causal reading. The body reports the attenuation as a question rather than a result because a floor effect—perturbation underperforming, compressing all three cells—would produce the same shape, and our one discriminating case points to it: on Styblinski-5D, where per-

Table S6: Exact-GP-minus-ensemble gap in raw oracle units (audited engine, X1 and X3 on, β=2, K=5, 30 seeds). "GP" is the exact GP throughout, matching the GP rows of Table S1; we use the single class rather than the better of the two per cell, since a per-cell maximum would be a selection statistic. Last column: perturbation gap as a percentage of the larger aggressive-optimizer gap. The negative entry on Ackley is the limiting case of the same pattern—there the ensemble edges the exact GP by 0.004 under perturbation, so the gap has not merely shrunk but closed.

Task	gradient	perturbation	CMA	pert. %
Branin-2D	6.95	0.053	6.71	0.8
Styblinski-5D	15.13	5.996	14.65	39.6
Levy-8D	0.46	0.047	0.53	8.8
Rosenbrock-10D	0.09	0.008	0.10	7.9
Rastrigin-15D	4.92	0.187	5.81	3.2
Ackley-20D	5.10	-0.004	5.32	-0.1
Griewank-30D	288.46	25.556	321.78	7.9

turbation is the grid's *best* optimizer (GP with perturbation at 35.9 raw), the attenuation is *weakest*, 39.6% of the larger aggressive-optimizer gap against at most 8.8% on the other six tasks, which is what a floor effect predicts. These raw-unit figures carry no normalization and no bootstrap.

Small-n bias in η^2. η^2 is positively biased at small n, and we disclose the size of that bias from our own artifacts rather than citing it in the abstract. Across the four corners the bootstrap mean exceeds the point estimate by $+0.0160$, $+0.0184$, $+0.0123$ and $+0.0099$ for the surrogate effect, which is the direction and rough magnitude the simulation literature reports (Okada 2013). Bias-correcting gives 0.351, 0.264, 0.438 and 0.395 against the reported 0.367, 0.283, 0.450 and 0.405. The two-confound comparison therefore moves $0.351 \rightarrow 0.395$ under correction, an increment of $+0.0437$ against the $+0.0376$ of the uncorrected pair: the direction survives and the magnitude of the change grows slightly. The interaction survives correction at 0.156, 0.134, 0.147 and 0.156. We name ϵ^2 rather than ω^2 as the correction because the simulation evidence identifies ϵ^2 as the least biased of the standard indices, against a common preference for ω^2 (Okada 2013).

An independent reproduction of the headline scalar. The β sweep and the corner decomposition are separate runs that meet at one operating point: both sit at X1 and X3 on, β=2, K=5, 30 seeds. They are genuinely independent rather than the same numbers twice—their per-cell means differ on 58 of the 63 cells, since the two runs draw different random streams—and they nonetheless give $\eta^2_{\text{surr}} = 0.40636$ from the β-sweep endpoint against 0.40455 from the headline corner, landing 0.0018 apart. For a scalar whose central challenge is that it is an operating-point artifact, an independent reproduction at the same operating point is worth stating explicitly rather than leaving a reader to read the two figures as one number rounded twice.

Table S7: Premise coverage at β=2 (synthetic mean), each surrogate on in-distribution points, its own proposals, and the other surrogate's proposals. Uncorrected engine (X1 off / X3 off), as in Table S1. "Other's proposals" means the points the *other surrogate's* pipeline returned, so this table crosses surrogate against proposal source, not against optimizer.

Premise	in-dist	own prop.	other's prop.
Ensemble	0.73	0.42	0.97
GP	0.98	0.97	0.93

6 Ablations

Pessimism strength β. Sweeping $\beta \in \{0, 0.5, 1, 2, 5\}$ (the β-resolution curve is Figure S6), increasing pessimism improves the final score on 6 of 7 synthetic tasks (median normalized slope $+0.19$); the exception is Ackley, where the uninformative σ makes the penalty point away from the optimum. That the benefit is distance-regularization rather than calibrated conservatism is confirmed by Figure S4: the per-task calibration quality ρ (rank correlation between σ and $|\mu - f|$) does not predict whether pessimism helps.

Ensemble size K. K is treated throughout as a confound to be controlled, never as a discovered mechanism. Sweeping $K \in \{2, 3, 5, 10\}$ on the audited engine (Figure S5), two facts are reported with equal weight. *Sensitivity:* η^2_{surr} is K-dependent—0.326, 0.366, 0.408, 0.389—peaking at exactly the K=5 the field fixes by convention, so the headline magnitude sits at the effect's maximum and we say so. *Non-reversal:* the ranking does *not* flip at K=2, where the ensemble marginal is 0.283 against the GP's 0.767. Our range $K \in \{2, 3, 5, 10\}$ shares three of its four points with the $K \in \{2, 5, 10\}$ over which ensemble surrogates were reported robust in online BO under a fixed acquisition rule (Li, Rudner, and Wilson 2024), adding only K=3; we claim no priority on the range, which already reached K=2. The residual is that the two sweeps measure different quantities: that work asks whether ensemble-BO *performance* is robust to K and finds that it is, while we ask whether the variance *attributed to the surrogate axis* is robust to K and find that it is not. Both can hold simultaneously, and only the second bears on a decomposition. Our pre-registered kill criterion for this sweep—that an ensemble marginal exceeding the GP's at K=2 would overturn the headline—did not fire.

The monotonically decreasing task-normalized score reported in an earlier draft of this supplement (0.95 at K=2) *does not reproduce* on the audited engine, which gives 0.283 at that cell. That figure is withdrawn, and no claim in this paper rests on it.

7 Official-Baseline Cross-Check

We ran the official `design-baselines` COMs and CbAS and compare to our reimplementations on the normalized scale (Table S8). CbAS matches closely (TF-Bind-8 $|\Delta|$=0.004); COMs diverges (TF-Bind-8 $|\Delta|$=1.22), reflecting a reduced-epoch official run and a different oracle variant. We therefore report both rather than a single number.

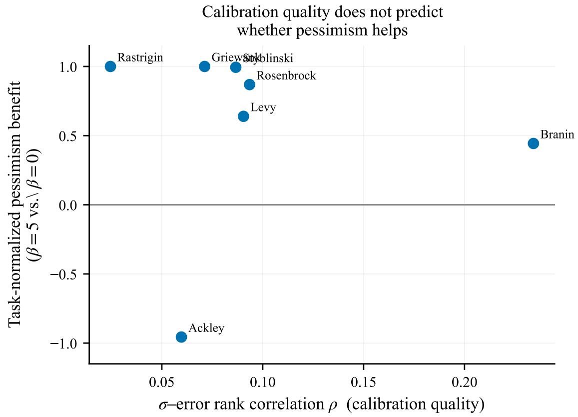

Figure S4: Calibration quality vs. pessimism benefit. Each point is a synthetic task; the σ–error rank correlation ρ (x-axis) does not predict the gain from pessimism (y-axis), consistent with a regularization rather than calibration effect.

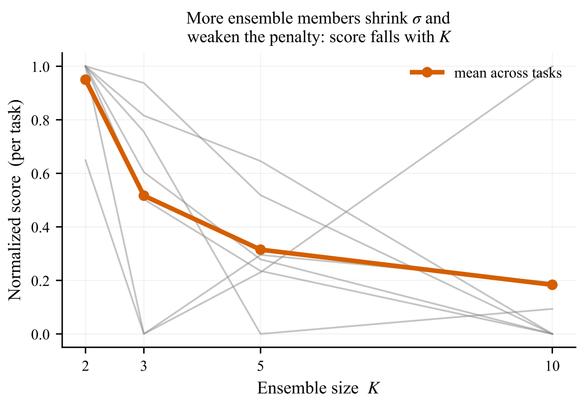

Figure S5: Ensemble-size ablation. Task-normalized score decreases with K.

8 Per-Task Premise Coverage

Table S9 gives the per-task one-sided coverage of the pessimism premise at β=2, with the fitted split-conformal multiplier $\hat{q}$ and the repaired coverages. In-distribution coverage is moderate-to-high and on several real tasks above nominal (the ensemble under-predicts there, giving negative $\hat{q}$); OOD coverage is task-dependent, near zero exactly on the tasks where gradient ascent collapses. Conformal restores in-distribution coverage to its 0.90 target on every task but leaves OOD coverage erratic. GFP is degenerate—coverage is measured on relaxed logits and is a decode artifact rather than a calibration signal.

9 Significance Details

Over the 9 grid cells and 7 tasks the synthetic Friedman omnibus gives p=6.1×10^{-5} on the *uncorrected* engine of Table S1 (Nemenyi critical difference 4.54; best cell GP+Grad, mean rank 2.29, bootstrap 95% CI $[1.29, 3.57]$ over 10,000 task resamples). On the audited engine the same omnibus gives p=8.1×10^{-4}, and every corner rejects: 6.1×10^{-5},

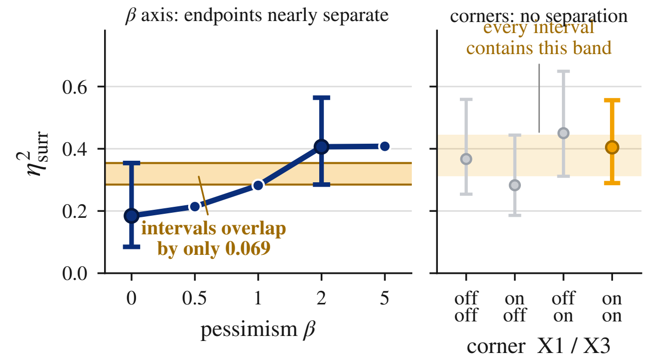

Figure S6: β nearly resolves where the engine corners do not (relocated from the main body, whose §4 prose reports every number here). Left: η^2_{surr} rises monotonically with pessimism, the β=0 and β=2 intervals overlapping by only 0.069 (intervals drawn at those two endpoints only). Right, on the same scale: the four corner intervals overlap completely. Synthetic grid, 7 tasks, 30 seeds, B=10,000.

Table S8: Official vs. reimplemented baselines (normalized 100th-percentile).

Method	Task	Official	Ours	$\lvert\Delta\rvert$
COMs	Superconductor	1.31	1.01	0.30
COMs	TF-Bind-8	0.99	2.21	1.22
COMs	GFP	−8.38	−9.20	0.82
CbAS	Superconductor	1.21	1.36	0.15
CbAS	TF-Bind-8	2.13	2.12	**0.004**
CbAS	GFP	2.20	1.85	0.35

1.7×10^{-3}, 4.1×10^{-5} and 8.1×10^{-4} for off/off, on/off, off/on and on/on respectively, as listed in Table S3. The conclusion is identical on either engine; we state the corner because the paper's own standard requires it. The Design-Bench omnibus over the same 9 cells fails to reject on the five-task set in every engine corner, p from 0.19 to 0.76 (best cell GP+Pert; the grid within one critical difference); adding the two MuJoCo tasks makes three of the four corners reject (p=0.011–0.014, off/off excepted at 0.16), which the body reports as an optimizer-axis effect under its own decomposition. Adding the domain baselines (COMs, CbAS, gradient ascent) to the five-task pool leaves the synthetic omnibus far below and the five-task Design-Bench omnibus far above α=0.05. Per-task Wilcoxon signed-rank tests with Holm correction are computed by the released `stats.py`; at the reported seed counts no Design-Bench pairwise comparison is significant after correction, and the five-task omnibus null is not rejected. As in the body, that is a failure to detect at this power and not a demonstration that an effect is absent (Agarwal et al. 2021): at n=7 a paired test requires $|d_z| \geq 1.27$ for 80% power at α=0.05, larger than anything we observe.

The equivalence bound, reported. The released code computes paired two-one-sided-tests, and rather than merely noting that they exist we give the result, because a bound is a more informative statement than a disclaimer. The

Table S9: One-sided coverage of the premise $\mu - f \leq \beta\sigma$ (i.e. $f \geq \mu - \beta\sigma$) at β=2 (nominal 0.90). $c_{\text{in}}/c_{\text{ood}}$: coverage in-distribution / on proposals; $\hat{q}$: signed split-conformal multiplier (negative $\Rightarrow$ the mean already over-covers); c^{cf}: coverage after replacing β with $\hat{q}$. Uncorrected engine (X1 off / X3 off), as in Table S1. On the audited engine the same measurement moves substantially—Branin's c_{in} is 0.82 rather than 0.71 and its c_{ood} 0.13 rather than 0.35—because the two corrections change what the optimizer proposes, which is exactly where OOD coverage is read. The qualitative pattern is the same on both engines and is the only thing we draw on: conformal repairs in-distribution coverage to 0.90 on every task and leaves OOD coverage erratic.

Task	c_{in}	c_{ood}	$\hat{q}$	$c^{\text{cf}}_{\text{in}}$	$c^{\text{cf}}_{\text{ood}}$
Branin	0.71	0.35	6.6	0.90	0.77
Styblinski	0.66	0.00	7.2	0.90	0.02
Levy	0.68	0.10	6.0	0.90	0.40
Rosenbrock	0.86	0.69	2.5	0.90	0.71
Rastrigin	0.73	0.79	4.8	0.90	0.83
Ackley	0.92	1.00	1.9	0.90	1.00
Griewank	0.57	0.00	15.9	0.90	0.07
TF-Bind-8	0.92	0.71	1.7	0.90	0.70
TF-Bind-10	1.00	0.51	−1.6	0.89	0.44
Superconductor	1.00	0.01	−2.1	0.90	0.00
GFP	0.00	0.00	67.4	0.90	1.00
UTR	1.00	0.00	−2.6	0.91	0.00
Ant	0.96	0.00	0.9	0.90	0.00
D'Kitty	0.53	0.00	7.7	0.90	0.00

best-versus-worst gap is 0.3762 with a 90% interval of $[-0.1078, 0.8602]$ and an effect bound of 0.4840. That is *not* equivalent at a 0.5 margin and not equivalent at a 0.3 margin either. So the Design-Bench null is a failure to detect that also fails to bound the effect below any margin one would call negligible—which is a positive statement about the null's reach, and a weaker one than equivalence in the direction that matters. Rejecting effects this small would require samples well beyond n=7 (Lakens 2017), and no claim in this paper rests on equivalence.

Robustness to the random-forest oracle. Because four of the seven Design-Bench tasks use a substituted random-forest oracle (GFP, UTR, Ant, D'Kitty), we check that the null is not manufactured by RF-flattened surfaces. Restricted to the three tasks whose oracle is *not* a smoothing RF substitution—TF-Bind-8/10 (exact oracles) and Superconductor (native random-forest oracle)—the Friedman omnibus over the 9 cells is p=0.93, and the median per-task method spread on the substituted tasks is *no smaller* than on these three—0.39 normalized over the three substituted tasks other than the degenerate GFP, 0.51 over all four, against 0.34 here—so the substitution did not compress the score range. The synthetic-to-real collapse of method separation is therefore a property of the real tasks, not an artifact of the oracle substitution.

Table S10: Normalizer robustness. Four engine corners × three per-task normalizers, recomputed from the stored per-seed grids. $\eta^2_{\text{surr}} > \eta^2_{\text{opt}}$ in all twelve; $\eta^2_{\text{inter}} > \eta^2_{\text{opt}}$ in eleven, inverting only in the marked row.

Corner	Norm.	η^2_{surr}	η^2_{opt}	η^2_{inter}	int/opt
off / off	min–max	0.367	0.013	0.165	12.4×
off / off	z-score	0.426	0.019	0.193	10.0×
off / off	rank	0.487	0.050	0.053	1.1×
on / off	min–max	0.283	0.036	0.146	4.0×
on / off	z-score	0.347	0.042	0.189	4.5×
on / off	rank	0.318	0.075	0.049	**0.6**×
off / on	min–max	0.450	0.006	0.152	26.9×
off / on	z-score	0.494	0.008	0.167	20.6×
off / on	rank	0.524	0.021	0.062	2.9×
on / on	min–max	0.405	0.005	0.160	32.8×
on / on	z-score	0.454	0.003	0.179	71.3×
on / on	rank	0.388	0.017	0.071	4.1×

10 Further Robustness Checks

All entries here are on the audited engine (X1 and X3 on, β=2, K=5, 30 synthetic seeds) unless stated.

Normalizer robustness. Recomputing the decomposition across the four engine corners and three per-task normalizers (Table S10), $\eta^2_{\text{surr}} > \eta^2_{\text{opt}}$ and η^2_{surr} is largest in *all twelve* combinations; the interaction exceeds the optimizer main effect in *eleven*, inverting only in the on/off corner under rank normalization (the corner carrying the largest optimizer effect). Rank transformation roughly halves the interaction throughout.

The three corrections to the prior analysis, in full. The body states these three; the counts are here. (i) *Gradient collapse.* Our released `gradtune.py` sweep concluded the ensemble's gradient collapse is an artifact a trust region repairs. By its own 5% rule the number of tasks whose collapse *survives* tuning is 2/7 with neither correction applied, 0/7 under target scaling alone, 5/7 under the candidate/oracle rule alone and 4/7 on the audited engine: genuine surrogate geometry on most tasks, driven by the candidate/oracle correction rather than by target scaling, and on the on/off corner not genuine at all. We report the corner that fires against us. (ii) *Not gradient-specific.* Gradient and CMA proposals differ in out-of-distribution coverage by 0.010 under the matched protocol against 0.117 under the unmatched one, because both optimizers pool every iterate and return the top-128 by surrogate LCB; our pre-registered X7 prediction of a gradient-specific effect is therefore retracted and the claim narrows to any aggressive selector. (iii) *Premise coverage.* The 0.97 of the prior analysis is the scikit-learn exact GP; the differentiable GP the grid actually scores averages 0.831 and falls to 0.38 on Styblinski-5D. Both are reported with the model each belongs to, and by Proposition 1 coverage of $\{\mu - f \leq \beta\sigma\}$ is exactly the lower bound's validity.

No landscape covariate predicts the gap. The GP–ensemble gap's rank correlation with dimension is +0.107 (gradient), +0.036 (perturbation) and +0.071 (CMA), indistinguishable from zero under every optimizer ($p \geq 0.81$) and pointing opposite the expectation that GP advantage decays with dimension (Li, Rudner, and Wilson 2024). Textbook multimodality mis-sorts: within the "many local minima" family the gap spans three orders of magnitude, while the one strictly unimodal task is among the smallest. Separability fares no better. This is a null from looking, not from failing to look, and not a reparametrization artifact—each task is mapped from the shared unit cube onto its exact native domain. One structural statement survives: the per-task gap *ordering* is stable across optimizers (Spearman 0.71–0.96; the gradient–perturbation pair at 0.71 is not significant at $n{=}7$, $p{=}0.07$) while the *magnitude* collapses under perturbation. Together these constrain any surviving mechanism to be multiplicative (a task factor scaled by an optimizer-aggressiveness factor); the collapse is bootstrapped, the ordering descriptive at $n{=}7$.

Far-field extrapolation does not distinguish the classes. A second pre-registered positive-mechanism arm raised the GP's prior mean to stop its far field reverting to the data mean. The manipulation does not move its target: a constant above every observation shifts the far-field posterior mean by under 1% of the distance to it ($R{=}0.009$ against a registered 0.25), because the fitted lengthscales are of order the domain diameter and no point in the feasible cube is in a reversion regime. The one arm that removes reversion carries $211\times$ the incumbent's held-out RMSE, and a surrogate that has stopped predicting is not evidence about reversion. Separately, probing the far field directly, the ensemble extrapolates near-linearly on all seven tasks while neither GP class reverts anywhere in the probed range, so far-field functional form does not distinguish the classes. A premise failed rather than a prediction surviving: any account of the GP advantage resting on prior reversion is unavailable here, not merely untested.

Three corrections to the budget measurement. The body reports the $11.8\times$ query-budget imbalance as *measured* rather than derived, and three corrections travel with that measurement. First, the previously estimated "6×–59×" spread was never measured, and CMA's nominal 3000 budget is a cap that rarely binds—pycma stops on its own tolerance criteria first, which is why the measured CMA median is 6,528 rather than 3000. Second, the proposal-count half of this confound is already removed by Confound 2, so the surviving inequality is in surrogate *queries* alone. Third, the trust region flagged in the audit is inactive in the main grid, so equalizing queries isolates budget rather than compounding two changes.

The sharpest Design-Bench budget objection. On Design-Bench the GFP CMA cell is where a budget confound could most plausibly manufacture the optimizer result: it spends 570 surrogate queries against gradient's $51{,}456$, a $90\times$ imbalance (at $d{=}4{,}740$ the library falls back to separable CMA and converges almost at once). Matching raises it to $51{,}494$, in line with the rest, and the seven-task and GFP-dropped sets move together, so repairing that starved cell did not by itself drive the outcome.

Table S11: Engine behind each supplementary result. "Uncorrected" is X1 off / X3 off; "audited" is X1 and X3 on. All synthetic entries are $\beta{=}2$, $K{=}5$ unless the item is itself a sweep over that coordinate.

Item	Engine	Seeds
Table S1 full grid	uncorrected	30
Table S2 average ranks	uncorrected	30
Table S3 four corners	all four corners	30
Matched tuning ($0.369 \to 0.275$)	uncorrected	30
$\beta{=}0$ controls, cross-check	audited	30
Matched subsample	uncorrected	30
Table S7 cross-proposal	uncorrected	30
Table S5 simple effects	all four corners	30
Table S6 raw-units gaps	audited	30
Small-n bias, ϵ^2	all four corners	30
β and K ablations	audited	30
Table S9 per-task coverage	uncorrected	30
Synthetic Friedman 6.1×10^{-5}	uncorrected	30
Design-Bench results	audited	16
33-combination coverage artifact	audited	8

11 Engine Map for Every Supplementary Number

The body's central methodological claim is that an η^2 without its operating point is not reproducible. That standard binds this supplement too, so Table S11 states the engine behind every number here. Two of the tables predate the corrections and are retained deliberately—they are what the body's corrections corrected—but retaining them without labelling them would be the exact failure this paper is about.

12 Experimental Configuration

Surrogates. Ensemble: $K{=}5$ MLPs, two hidden layers of width 96, ReLU, MSE, 35 epochs, Adam (lr $3{\times}10^{-3}$, weight decay 10^{-4}), batch 256, per-member seed offset. Exact GP: differentiable single-task GP, ARD Matérn-$5/2$, marginal-likelihood fit (frozen under matched tuning), score-biased subsample to $N_{\max}{=}800$. SVGP: 128 inducing points, ARD Matérn-$5/2$, 250 variational-ELBO steps (Adam lr 0.01), $N_{\max}{=}2000$.

Optimizers. Gradient: 100 Adam steps on x (lr 0.05), box-clipped. Perturbation: 5 rounds, $\sigma \in \{0.1, 0.05, 0.02\}$, elementwise accept-if-better. CMA-ES: initial $\sigma{=}0.2$, bounds $[0, 1]$, separable (diagonal) covariance when $d{>}500$.

Acquisition and protocol. LCB $\mu - \beta\sigma$, $\beta{=}2$. Candidate budget 128 (top-128 dataset points plus $\sigma{=}0.05$ perturbations as starts; top-128 by surrogate score evaluated by the oracle). Metrics: 100th- and 50th-percentile normalized scores. Seeds: 30 synthetic, 16 Design-Bench. The β sweep, the K sweep and the per-task calibration measurement are also 30 seeds each, on the audited engine; the 33-combination coverage artifact is 8.

Tasks. Synthetic: Branin-2D ($N{=}2000$), Styblinski-5D (3000), Levy-8D (4000), Rosenbrock-10D (5000), Rastrigin-15D (5000), Ackley-20D (5000), Griewank-30D (8000);

dataset drawn once at seed 0 and fixed across seeds. Design-Bench: TF-Bind-8/10 (exact oracles), Superconductor, GFP, UTR, Ant, D'Kitty (random-forest oracles); discrete designs relaxed to per-position class logits, decoded by argmax; inputs and scores min-max normalized; raw rows score-biased-subsampled to 8000 before encoding.

Compute and environment. Every run is *CPU-only*: each stamped artifact carries a CPU build of PyTorch (2.8.0+cpu or 2.11.0+cpu), and no cell in this paper touches a GPU or a CUDA kernel. The synthetic grid and the Design-Bench grid were produced on x86-64 Linux (kernel 6.17.0-35-generic, glibc 2.39) under Python 3.12.3 for the synthetic environment and 3.9.23 for the Design-Bench environment, which pins the older library stack that `design-bench` requires; the single exception is the TF-Bind-8 platform check, run on arm64 macOS 26.3.1. Each result file records its own `platform`, `os_release`, `python`, `torch`, `numpy`, `botorch`, `gpytorch`, `cma` and `git_sha` beside the engine coordinates, and the analysis loader refuses any file missing one of them, so no number here can be read apart from the environment that produced it. Cells are independent (task, variant, seed) triples dispatched to a process pool of one worker per core less one; the recorded preview run cleared 100 of 312 cells in 8.4 minutes on eight workers, so at that rate the 1,890-cell synthetic grid is a matter of hours rather than days on a commodity multi-core machine. The work is CPU-bound and embarrassingly parallel, with no distributed scheduling and no accelerator dependency to reproduce.

Calibration measurement. One-sided premise coverage $\widehat{\Pr}(\mu - f \leq \beta\sigma)$ (the LCB lower-bound event $f \geq \mu - \beta\sigma$) estimated in-distribution on uniform test points and out-of-distribution on the gradient-ascent proposals, at $\beta \in \{0.5, 1, 2, 5\}$. Split-conformal multiplier $\hat{q}$ from the signed nonconformity $(\mu - f)/\sigma$ with a relative σ floor (5% of the calibration-fold mean), fit at δ=0.1 on a held-out fold; $\rho_{\text{err}} = \text{Spearman}(\sigma, |\mu - f|)$.

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